

Virtualisation and Open Source

What Makes It the Right Match?

Virtualisation continues to be a buzzword. The idea isn't new in itself; IBM mainframes have always had to have a hypervisor. It's in the news now because even simple desktops can now act as virtual machine hosts. A lot of possibilities have opened up as a result. Let's take a brief tour of what virtualisation means, in its classic sense, and look at why open source virtualisation is going to win.

Well, we all know what virtualisation is about—we've read and heard of it over and over again. But let's look at it from the view point of the open source world.

Virtualisation means the simulation of a computer system, in software. The virtualisation software creates an environment for a guest, a complete OS, to execute within this created world. This means the view that should get exported to the guest should be of a complete computer system—with the processor, system peripherals, devices, buses, memory and so on. The virtualisation software can be strict about what view to export to the guest, for example, the processor and processor features, types of devices, buses exported to the guest, etc, or it can be flexible with the user getting a choice to select individual components and parameters.

There are some constraints to creating a virtualised environment or a set of sufficient requirements, as has been noted by Popek and

Goldberg in their paper on virtual machine monitors.

- **Fidelity:** Software running in a virtualised environment should not be able to detect it is running on a virtualised system.
- **Containment:** Activities within a virtual machine (VM) should be contained within the VM itself without disturbing the host system. A guest should not cause the host or other guests running on the host to malfunction.
- **Performance:** Performance is crucial to how the user sees the utility of the virtualising environment. In this age of extremely fast and affordable general-purpose computer systems, if it takes a few seconds for some input action to get registered in a guest, no one will be interested in using the virtual machine at all.
- **Stability:** The virtualisation software itself should be stable enough to handle the guest OS and any quirks it may exhibit. There are several reasons why one would